# Homework 08

Haoran Xu

2026-10-18

All work is in base R. No classifier is fitted anywhere: Questions 1 and 2 only evaluate the score functions given in the questions.

## Question 1 (Boundary and margin)

### Solution

**Part a.** With $w = (0.8, 0.6)$ and $\|w\|_2 = 1$, the signed distance from $x$ to the boundary $f(x) = w^\top x = 0$ is $f(x)/\|w\| = f(x)$, so the (unsigned) distance is $|f(x)|$ and the functional margin is $y\,f(x)$. An observation is *on* the margin when $yf(x) = 1$, *inside* when $yf(x) < 1$ (this includes the misclassified points, where $yf(x) < 0$), and *beyond* when $yf(x) > 1$.

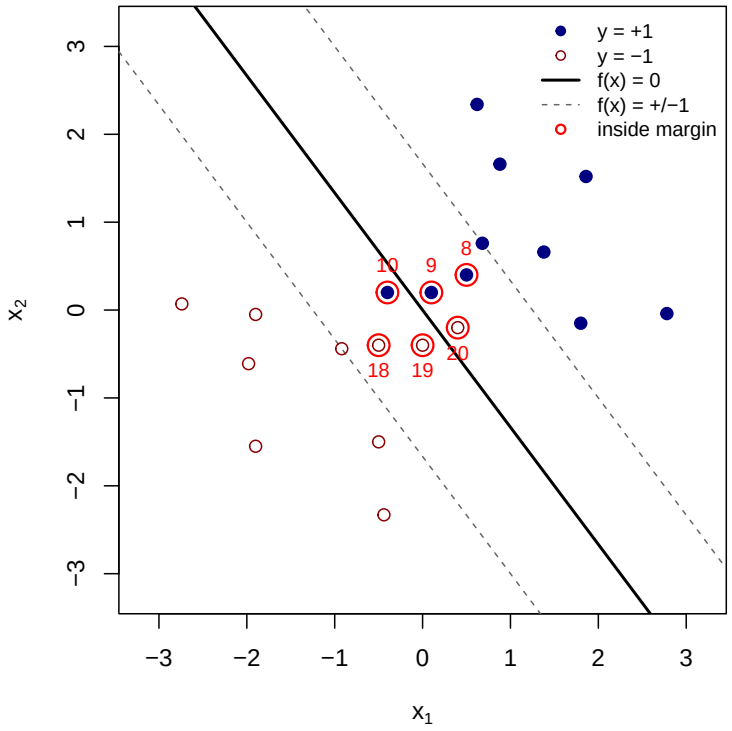
```
d <- read.csv("data/margin-geometry-20.csv")
d$f      <- 0.8 * d$x1 + 0.6 * d$x2          # score
d$dist   <- abs(d$f)                         # distance to f(x) = 0 (||w|| = 1)
d$margin <- d$y * d$f                        # functional margin y f(x)
d$status <- ifelse(abs(d$margin - 1) < 1e-8, "on",
                   ifelse(d$margin < 1, "inside", "beyond"))
d$correct <- ifelse(d$margin > 0, "yes", "no")
knitr::kable(d, digits = 2, align = "r",
             col.names = c("obs", "$x_1$", "$x_2$", "$y$", "$f(x)$",
                           "distance", "$y f(x)$", "margin status", "correct"))
```

| obs | $x_1$ | $x_2$ | $y$ | $f(x)$ | distance | $yf(x)$ | margin status | correct |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.68 | 0.76 | 1 | 1.00 | 1.00 | 1.00 | on | yes |
| 2 | 1.80 | -0.15 | 1 | 1.35 | 1.35 | 1.35 | beyond | yes |
| 3 | 1.38 | 0.66 | 1 | 1.50 | 1.50 | 1.50 | beyond | yes |
| 4 | 0.88 | 1.66 | 1 | 1.70 | 1.70 | 1.70 | beyond | yes |
| 5 | 0.62 | 2.34 | 1 | 1.90 | 1.90 | 1.90 | beyond | yes |
| 6 | 2.78 | -0.04 | 1 | 2.20 | 2.20 | 2.20 | beyond | yes |
| 7 | 1.86 | 1.52 | 1 | 2.40 | 2.40 | 2.40 | beyond | yes |
| 8 | 0.50 | 0.40 | 1 | 0.64 | 0.64 | 0.64 | inside | yes |
| 9 | 0.10 | 0.20 | 1 | 0.20 | 0.20 | 0.20 | inside | yes |
| 10 | -0.40 | 0.20 | 1 | -0.20 | 0.20 | -0.20 | inside | no |
| 11 | -0.92 | -0.44 | -1 | -1.00 | 1.00 | 1.00 | on | yes |
| 12 | -0.50 | -1.50 | -1 | -1.30 | 1.30 | 1.30 | beyond | yes |
| 13 | -1.90 | -0.05 | -1 | -1.55 | 1.55 | 1.55 | beyond | yes |
| 14 | -0.44 | -2.33 | -1 | -1.75 | 1.75 | 1.75 | beyond | yes |
| 15 | -1.98 | -0.61 | -1 | -1.95 | 1.95 | 1.95 | beyond | yes |
| 16 | -2.74 | 0.07 | -1 | -2.15 | 2.15 | 2.15 | beyond | yes |
| 17 | -1.90 | -1.55 | -1 | -2.45 | 2.45 | 2.45 | beyond | yes |
| 18 | -0.50 | -0.40 | -1 | -0.64 | 0.64 | 0.64 | inside | yes |
| 19 | 0.00 | -0.40 | -1 | -0.24 | 0.24 | 0.24 | inside | yes |

| obs | $x_1$ | $x_2$ | $y$ | $f(x)$ | distance | $yf(x)$ | margin status | correct |
|---|---|---|---|---|---|---|---|---|
| 20 | 0.40 | -0.20 | -1 | 0.20 | 0.20 | -0.20 | inside | no |

**Part b.** The boundary is $0.8x_1 + 0.6x_2 = 0$, i.e. $x_2 = -\frac{4}{3}x_1$, and the margin lines $f(x) = \pm 1$ are $x_2 = -\frac{4}{3}x_1 \pm \frac{5}{3}$. Filled points are class $+1$, hollow points class $-1$; the six observations inside the margin are circled in red and labelled.

```
par(mar = c(4, 4, 1, 1), pty = "s")
plot(NA, xlim = c(-3.2, 3.2), ylim = c(-3, 3), asp = 1,
     xlab = expression(x[1]), ylab = expression(x[2]))
abline(0, -4/3, lwd = 2)                          # f(x) = 0
abline( 5/3, -4/3, lty = 2, col = "grey40")  # f(x) = +1
abline(-5/3, -4/3, lty = 2, col = "grey40")  # f(x) = -1
points(d$x1, d$x2, pch = ifelse(d$y == 1, 19, 1), col = ifelse(d$y == 1, "navy", "darkred"),
       cex = 1.1)
ins <- d$status == "inside"
points(d$x1[ins], d$x2[ins], pch = 21, cex = 2, col = "red", lwd = 1.5)
text(d$x1[ins], d$x2[ins], d$observation[ins], col = "red", cex = 0.8,
     pos = ifelse(d$y[ins] == 1, 3, 1), offset = 0.7)   # +1 labels above, -1 below
legend("topright", bty = "n", cex = 0.85,
       legend = c("y = +1", "y = -1", "f(x) = 0", "f(x) = +/-1", "inside margin"),
       pch = c(19, 1, NA, NA, 21), lty = c(NA, NA, 1, 2, NA), lwd = c(NA, NA, 2, 1, 1.5),
       col = c("navy", "darkred", "black", "grey40", "red"))
```




**Part c.**

```
table(factor(d$status, levels = c("inside", "on", "beyond")))
#>
#> inside     on beyond
#>      6      2     12
d$observation[d$margin < 0]                  # misclassified: y and f(x) disagree
#> [1] 10 20
```

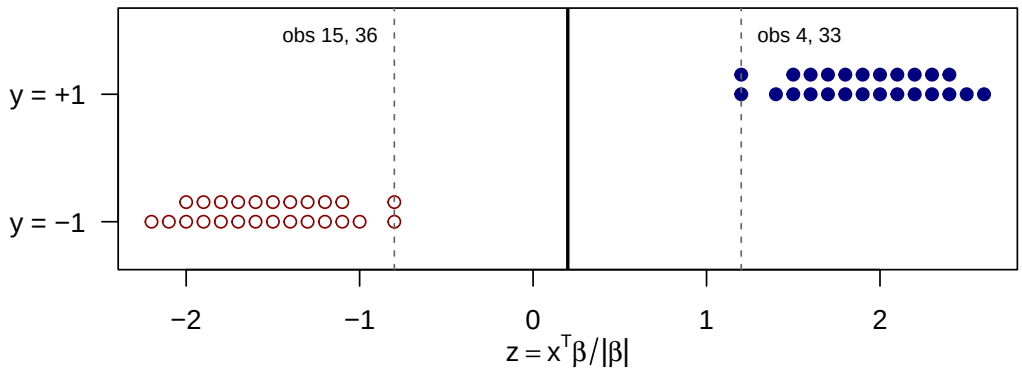
Twelve observations lie beyond the margin, two lie exactly on it (observation 1 with $f = 1$ and observation 11 with $f = -1$), and six lie inside it (observations 8, 9, 10, 18, 19, 20). Two of the six are misclassified: observation 10 (class $+1$, $f = -0.2$) and observation 20 (class $-1$, $f = 0.2$), both with functional margin $-0.2$. The other four inside points (8, 9, 18, 19) are on the correct side but closer than distance 1 to the boundary. With this fixed rule the training error is $2/20 = 0.1$.

## Question 2 (Margin in four dimensions)

### Solution

**Part a.** $\|\beta\|_2 = \sqrt{1+4+4+16} = 5$, so the unit direction is $\mathbf{u} = \beta/5 = (0.2, 0.4, -0.4, 0.8)^\top$. Projecting an observation onto $\mathbf{u}$ gives the scalar $z_i = \mathbf{x}_i^\top \mathbf{u}$, and the score is $f(\mathbf{x}_i) = \beta_0 + 5z_i$. The boundary $f = 0$ therefore sits at $z^\star = -\beta_0/5 = 0.2$ on the projection axis, and the signed distance of observation $i$ from the boundary is $z_i - z^\star = f(\mathbf{x}_i)/5$.

```
d4 <- read.csv("data/margin-geometry-4d.csv")
beta <- c(1, 2, -2, 4); beta0 <- -1
X <- as.matrix(d4[, c("x1", "x2", "x3", "x4")])
f  <- as.vector(beta0 + X %*% beta)
z  <- as.vector(X %*% beta) / sqrt(sum(beta^2))    # position along beta / ||beta||
zb <- -beta0 / sqrt(sum(beta^2))                    # boundary on the axis
par(mar = c(3.5, 3.5, 0.5, 1))
stripchart(list("y = -1" = z[d4$y == -1], "y = +1" = z[d4$y == 1]),    # stack ties
           method = "stack", offset = 0.6, pch = c(1, 19), col = c("darkred", "navy"),
           las = 1, xlab = "", ylim = c(0.7, 2.6), xlim = range(z))
mtext(expression(z == x^T * beta / group("||", beta, "||")), side = 1, line = 2.3)
abline(v = zb, lwd = 2); abline(v = zb + c(-1, 1), lty = 2, col = "grey40")
close <- which(abs(abs(z - zb) - min(abs(z - zb))) < 1e-8)    # nearest the boundary
for (k in c(-1, 1)) text(zb + k, 2.45, cex = 0.75, pos = 3 + k,    # name the closest points
     paste("obs", paste(d4$observation[close][d4$y[close] == k], collapse = ", ")))
```




```
all(d4$y * f > 0)                    # every observation on its own side
#> [1] TRUE
tapply(z - zb, d4$y, function(v) range(abs(v)))    # distances to boundary by class
#> $`-1`
#> [1] 1.0 2.4
#>
#> $`1`
#> [1] 1.0 2.4
M <- min(abs(z - zb)); M
#> [1] 1
d4$observation[abs(abs(z - zb) - M) < 1e-8]         # observations attaining M
#> [1]  4 15 33 36
```

Every observation is on the correct side of the boundary, so the data are linearly separated by this rule. The class $-1$ points occupy $z \in [-2.2, -0.8]$ and the class $+1$ points $z \in [1.2, 2.6]$, so the nearest points on each side are exactly one unit from the boundary at $z^\star = 0.2$. The geometric margin is $M = 1$, attained by four observations (two from each class, named in the plot), and the margin band $z^\star \pm M$ contains no observations.

**Part b.** The smallest functional margin under the given parameters is $\min_i y_i f(\mathbf{x}_i) = 5$ (equivalently $M\|\beta\| = 1 \times 5$). Rescaling $(\beta_0, \beta)$ by any positive constant leaves the boundary unchanged, so dividing by 5 gives the SVM normalisation:

```
m <- min(d4$y * f); m
#> [1] 5
beta_svm <- beta / m; beta0_svm <- beta0 / m
beta_svm; beta0_svm
#> [1]  0.2  0.4 -0.4  0.8
#> [1] -0.2
min(d4$y * (beta0_svm + X %*% beta_svm))   # smallest functional margin is now 1
#> [1] 1
1 / sqrt(sum(beta_svm^2))                  # geometric margin = 1 / ||beta_svm||
#> [1] 1
```

$$\beta_{\text{svm}} = \tfrac{1}{5}(1, 2, -2, 4)^\top = (0.2, 0.4, -0.4, 0.8)^\top, \qquad \beta_{0,\text{svm}} = -\tfrac{1}{5} = -0.2.$$

The margin points now satisfy $y_i f(\mathbf{x}_i) = 1$ exactly, and $1/\|\beta_{\text{svm}}\| = 1 = M$, consistent with part (a).

## Question 3 (Peer review of a Homework 7 skill)

### Solution

**Assignment.** The instructor's issue (rqzhu-aide, issue #2 in my repository) assigned the skill from `zexuanj/stat432-fall2026`, `homework/week-07/SKILL.md`. The file (3,558 bytes, skill name `analyze-lda-qda`) was copied unchanged to `homework/week-08/peer-review/SKILL.md`. It was not installed: its own `description` says to load it only on an explicit request to read the file, so I passed the path in the prompt.

**Part 1: using the skill.** The agent (Claude Code, run as a background subagent) received the prompt below and nothing else. The only non-data instructions are where to save files, which language and tools exist on the machine, and not to open solution files.

```
Read the skill file at homework/week-08/peer-review/SKILL.md first and follow
that skill exactly while analyzing the SAheart data.
Data information:
- Dataset: homework/week-08/data/saheart.csv (462 rows, 10 columns).
- Response: `chd` (0 = no coronary heart disease, 1 = CHD).
- Predictors: the other nine columns (sbp, tobacco, ldl, adiposity, famhist,
  typea, obesity, alcohol, age), with `famhist` coded numerically as
  1 = Absent, 2 = Present.
- Random seed: 432.
Work in the folder homework/week-08/peer-review/ and save every file you create
there (code, the report, any rendered output). Use R (MASS is installed) and
Quarto, which are available on this machine. Do not open or read any file whose
name contains "solutions", and do not modify anything outside that folder.
When finished, report back: the list of files you produced, a plain summary of
what the analysis did and found (candidate models, selected model, CV and test
errors, confusion matrix), any places where the skill was ambiguous, unclear,
or impossible to follow as written and what you did in each case, and whether
you verified that the rendered report fits on one page.
```

*Monitoring.* The run was watched from the parent session while Questions 1 and 2 were being written. It started at 20:55:52 and finished at 21:01:48 (5 min 56 s wall time), used 64,861 tokens and 16 tool calls, and finished on its own; no steer message was needed and the run was not stopped early. Weekly quota use was negligible (well under 1% of the allowance).

*What the skill produced.* Files in `peer-review/`: `analysis.R` (9.2 KB, `MASS` only), `report.qmd` and `report.pdf` (one page, reproduced on the next page without edits), and a `results/` folder with the split indices, fold assignments, all 18 fold-error rows, a per-fold log (rank check, condition numbers, warnings), the candidate table, and an `.rds` with everything including `sessionInfo()`. The pipeline followed the skill's eight steps: stratified 30% test set under seed 432 (139 test / 323 train), five stratified folds fixed once, 18 candidates (ordinary LDA, ordinary QDA, standardized PCA-LDA and PCA-QDA for 1 to 8 components), all preprocessing fitted on the analysis rows of each fold, a within-class rank check in every fold (nothing rejected, no warnings), selection by minimum mean CV error, one refit and one test evaluation, with the two baselines scored at that same stage.

| Model | Dims | Mean CV error (SE) | Test error (binomial SE) |
|---|---|---|---|
| Ordinary LDA | 9 | 0.266 (0.007) | 0.302 (0.039) |
| Ordinary QDA | 9 | 0.287 (0.036) | 0.317 (0.039) |
| Selected: PCA-LDA (5) | 5 | 0.254 (0.016) | 0.302 (0.039) |

The selected model, PCA-LDA with 5 components, made 42 errors on the 139 test rows (observed 0: 76 right, 15 wrong; observed 1: 21 right, 27 wrong), the same count as ordinary LDA. The agent reported eight places where the skill was ambiguous and how it resolved each: (i) `famhist` is stored as text, so it recoded 1/2 as instructed; (ii) "one figure and one table" conflicts with the required confusion matrix, so the matrix went into prose; (iii) "dimensions" is undefined for the raw-predictor models, taken as 9; (iv) every PCA dimension was "feasible", so none was dropped; (v) "settle skill revisions" is not actionable in a single run; (vi) the 30% stratified split does not say how to round per class; (vii) axis labels are slightly below 10 pt, treated as not "body text"; (viii) a general plotting skill auto-loaded and its colour guidance was applied to the figure. I reran `analysis.R` and the numbers in the table match the script output.

**Part 2: peer review.** The review below was posted verbatim as the GitHub Issue in Part 3.

*Overall assessment.*

The skill's protocol is sound: a stratified test set reserved first, fixed folds reused for every candidate, fold-local preprocessing, a rank check for QDA, and one test evaluation with the baselines scored only at that stage. It ran cleanly on SAheart with seed 432 and produced a leak-free pipeline, a one-page report, and saved split and fold results. The weaknesses are inconsistent reported numbers, a greedy selection rule, and a few ambiguous instructions.

*Item 1: the reported results contradict each other.*

**Observed.** The written summary says PCA-QDA(5) was selected (CV 0.2385, test 0.2950; LDA 0.2662, QDA 0.2806). The embedded one-page report, same seed and split, says PCA-LDA(7) (CV 0.272, test 0.302; LDA 0.302, QDA 0.309). No script or saved results are attached.

**Why it matters.** Two different "final" results from one seed means the prose and the page come from different runs, which defeats the single-test-evaluation design and makes the analysis unreproducible.

**Recommendation.** Regenerate both from one script run, attach the script, and require in the skill that the report read every number from the saved results file.

*Item 2: the minimum-CV rule is too greedy for this sample size.*

**Observed.** Eighteen candidates are compared, fold SEs are 0.01 to 0.04, and the gaps between the leaders are about 0.01. The author's run chose PCA-QDA(5), which then did worse on test than both baselines; my run chose PCA-LDA(5), which tied ordinary LDA on test. The figure's 0 to 0.4 y-axis hides these small differences.

**Why it matters.** Picking the minimum of 18 noisy estimates mostly selects noise, and the skill treats selection uncertainty only as a limitation to mention.

**Recommendation.** Add a one-SE rule anchored at ordinary LDA, require one sentence on whether the winning margin exceeds one SE, and fit the figure's y-range to the data.

*Item 3: ambiguous instructions and missing required material.*

**Observed.** "One figure and one table" conflicts with the required confusion matrix; my run put the matrix in prose, while the author's own report shows two tables. Also undefined: the coding of the text-valued `famhist`, the "dimension" of raw LDA/QDA, and per-class rounding of the 30% split. The submission omits the full `SKILL.md` text, gives no account of checking or revising the agent's work, and pastes the same prompt twice.

**Why it matters.** Each ambiguity makes two runs of the same skill diverge, and without the skill text or a review

step a reader cannot verify what the agent was told or whether leakage was checked.

**Recommendation.** Resolve the table rule explicitly, define the coding and dimensions, append the skill text, remove the duplicate prompt, and add a short paragraph on what was checked in the agent's script.

**Part 3: link to the GitHub Issue.** https://github.com/zexuanj/stat432-fall2026/issues/4 (issue #4, "homework 8 review", opened in `zexuanj/stat432-fall2026` on 2026-10-18).

## Appendix: the one-page report produced by the assigned skill (unedited)

### LDA and QDA for coronary heart disease (SAheart)

Haoran Xu

**Data and coding.** The SAheart data have 462 men and 9 predictors (sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age); the response `chd` is 0 = no coronary heart disease (302) or 1 = CHD (160). `famhist` arrived as "Absent"/"Present" and was coded 1 = Absent (270), 2 = Present (192), so it is a binary predictor; tobacco and alcohol are right-skewed with many zeros. There were no missing values and no identifier columns.

**Design.** With seed 432, a stratified 30% test set (139 rows: 91 / 48) was reserved before any modelling; 323 rows (211 / 112) remained for training, and were split once into five stratified folds reused by every candidate. Candidates were ordinary LDA (pooled covariance), ordinary QDA (one covariance per class), and standardized PCA followed by LDA or QDA with 1–8 retained components (18 models). Within each fold, the standardization, PCA rotation, class means, empirical priors and covariances were estimated from the four analysis folds only and applied to the held-out fold. Within-class rank was checked after preprocessing in every fold; all 18 candidates had full rank in all five folds, with no numerical warnings. The selection score was the mean validation misclassification rate (SE = SD of fold errors / $\sqrt{5}$), ties broken by fewer dimensions and then LDA. Only after selection was the chosen pipeline refit on all 323 training rows and scored once on the test set; the two ordinary baselines were scored at the same stage for comparison only.

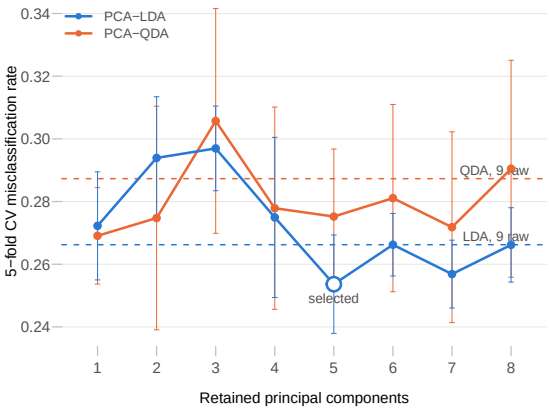



Table 1: Mean 5-fold CV error (SE) on n = 323 and one test evaluation (binomial SE) on n = 139.

| Model | Dims | Mean CV (SE) | Test (SE) |
|---|---|---|---|
| Ordinary LDA | 9 | 0.266 (0.007) | 0.302 (0.039) |
| Ordinary QDA | 9 | 0.287 (0.036) | 0.317 (0.039) |
| Selected PCA-LDA (5) | 5 | 0.254 (0.016) | 0.302 (0.039) |

**Figure.** Mean CV error (±1 SE) against retained dimension for PCA-LDA (blue) and PCA-QDA (orange); dashed lines are the ordinary 9-predictor LDA and QDA. **Selection.** The minimum mean CV error was PCA-LDA (5) at 0.254 (SE 0.016), ahead of PCA-LDA (7) at 0.257 and ordinary LDA at 0.266 (SE 0.007); every QDA variant was worse (ordinary QDA 0.287, SE 0.036) and showed much larger fold-to-fold variability. The differences among the leading LDA candidates are within about one SE, so the choice of 5 components is a minimum-CV rule, not strong evidence that 5 is the "right" dimension.

**Test result for the selected model (PCA-LDA, 5 components).** Rows are observed classes, columns predicted: observed 0 (n = 91): 76 predicted 0, 15 predicted 1; observed 1 (n = 48): 27 predicted 0, 21 predicted 1 (totals 139 = test size). Test errors: 42/139 = 30.2%, with approximate binomial SE $\sqrt{\hat{p}(1-\hat{p})/139} = 3.9\%$. Ordinary LDA made the same number of test errors (30.2%) and ordinary QDA 31.7%; all three are indistinguishable at this test size. The model is far better at identifying non-cases (15/91 = 16.5% false-positive rate) than cases (27/48 = 56.2% missed), a consequence of the empirical priors (about 65% / 35%) and heavily overlapping classes.

**Interpretation and limitations.** CHD risk in these data is mostly a shared linear shift (age, tobacco, ldl, famhist, typea) rather than class-specific covariance structure, so the extra parameters of QDA buy nothing and add variance; PCA reduction mainly stabilizes the LDA estimates. The test error (30.2%) is higher than the CV estimate (25.4%), which is expected because the CV figure is a minimum over 18 candidates and is optimistically biased as a performance estimate; the gap is also within one test SE. No candidate was singular, but the raw-scale covariances have condition numbers near 3,189 (LDA) and 4,307 (QDA) purely because predictors are on very different scales (sbp in the hundreds, famhist in {1, 2}); the standardized PCA pipelines had condition numbers below 9. The Gaussian assumption is clearly violated by the binary `famhist` and the zero-inflated, skewed tobacco and alcohol, which hurts QDA most. The test set holds only 139 men (48 cases), giving a test SE near 4 percentage points, and the result depends on one seed and one fold assignment. The selected model is the minimum-CV choice among these LDA/QDA families for this split, not a universally optimal classifier.